Energy-Audited Lagrangian Simulation and Sensitivity-Gradient Tipping for Cable-Only Locomotion of a Six-Bar Tensegrity Rover

[TODO: Author names]
[TODO: Department of Robotics, Institution, City, Country]
[TODO: email]

***Abstract*—Simulation results for rolling tensegrity robots are hard to trust when the simulator itself can create or destroy energy: penalty-spring bars followed by position projection, ground clamps and "helper" torques can all produce apparent locomotion that the cables alone could not. We present a simulator and a gait for a six-bar tensegrity rover with an internal payload in which every joule is accounted for. The shell is modelled in non-minimal Cartesian coordinates with the consistent rigid-bar mass matrix and exact bar-length multipliers; contact is a passive Hunt–Crossley law whose elastic part is a potential; and a work ledger books every force and every projection and measures, at each step and before any correction, the energy that no booked work explains. This per-step drift is first-order discretization error (it halves when the time step is halved) and is then removed by a minimum-norm constraint-and-energy corrector. The audit shows that a conventional penalty+SHAKE engine exchanges 6.8×10^3 J with its position projection to keep the bars rigid and, in its interactive form, takes 50 % of its energy input from non-physical aids. On the audited simulator we propose Sensitivity-Gradient Tipping (SGT), which differentiates the pinned quasi-static equilibrium, including the softening geometric stiffness of the compressed bars, to obtain the sensitivity of the centre of mass to all 36 shell and payload cables, and drives it across a forward support edge within the actuator box. With cables only, SGT reaches 0.091 m/s (100 % success over 5 seeds), whereas reduced-order LQR, iLQR and MPC schedulers over a support-face primitive advance at most 0.10 m in 20 s under identical limits. Ablations show that the geometric-stiffness term is the component that matters, and stress tests expose a sensitivity to heavy-tailed sensor noise. Code, run files and the scripts that produce every number are released.**

***Index Terms*—Tensegrity robots, legged and rolling locomotion, simulation and animation, underactuated robots, energy-consistent integration.**

I. Introduction

Six-bar spherical tensegrities combine low mass with high impact tolerance and have been studied as planetary landers and rovers [1], [2]. They move by *punctuated rolling*: changing cable rest lengths deforms the shell until the centre of mass (COM) leaves the triangular support face and the structure tips onto a neighbouring face. Because hardware is expensive and fragile, gaits are usually developed in simulation [3], [4] and transferred later.

Two practical problems limit what such simulations prove. *(i) Admissibility.* Rigid bars are often emulated by stiff penalty springs and corrected by position projection (SHAKE/RATTLE [5], [6]); contacts by clamps or restitution impulses. Each of these steps changes the mechanical energy without a corresponding physical work term, and in interactive simulators it is common to add "assist" torques or traction forces so that a gait looks convincing. Without an audit it is impossible to tell how much of the locomotion the cables actually produced. *(ii) Actuation.* Classic support-face gaits contract a fixed set of cables around the current face. With a realistic actuator stroke and a heavy internal payload, such a primitive may not move the COM far enough to tip, regardless of how its effort is scheduled.

This paper addresses both problems.

1) *Energy-audited simulation.* We implement the non-minimal Cartesian Lagrangian formulation of tensegrity dynamics with the consistent rigid-bar mass matrix, exact constraint multipliers and a minimum-norm constraint-and-energy corrector [7], and add a passive Hunt–Crossley contact [8] whose elastic energy is part of the energy function. A *work ledger* books every force category and every projection and records the per-step energy mismatch *before* correction, which bounds any unbooked energy source and can be checked for convergence in Δt.
2) *Sensitivity-Gradient Tipping (SGT).* A cable-only gait that linearizes the quasi-static equilibrium of shell and payload with the support pinned, including the geometric stiffness of the compressed bars, and commands all 36 actuators along the gradient of the COM displacement across a forward support edge.
3) *Protocol and audit.* A deterministic, seed-locked benchmark shared by the interactive application and a headless batch runner: five baselines with identical actuator limits, impulse and step disturbances, Gaussian/Laplace/outlier sensor noise, stiffness/friction/mass mismatch and actuation delay, seven ablations and three terrains (114 runs). Tables, figures and every number in the text are

generated from the run files.

To our knowledge, the combination of a per-step energy audit with a sensitivity-based cable-only tipping gait has not been reported for six-bar tensegrity rovers. All claims are restricted to simulation (Section VII); in particular the negative results for the baselines concern the support-face primitive they schedule, not the optimal-control methods as such.

II. Related Work

Tensegrity simulation. General engines such as MuJoCo and Bullet [9], [10] and dedicated tensegrity simulators [3] treat bars as rigid bodies or stiff springs; constraint drift is removed by stabilization [11] or projection [5], [6]. Structure-preserving integrators [12] conserve energy for conservative systems but do not by themselves expose energy injected by contact or by non-physical controller aids. The class-k tensegrity literature [13] and the non-minimal Cartesian formulation [7] provide the dynamics used here.

Tensegrity locomotion. Rolling gaits have been obtained from hand-designed cable sequences, kinodynamic planning over gait primitives [2] and reinforcement learning [4]. These works optimize *which* primitive to execute; SGT instead computes the primitive itself from the local quasi-static sensitivity of the COM and is therefore complementary to primitive-level planners.

III. Energy-Audited Shell Dynamics

A. Coordinates, mass and constraints

The shell has $n = 12$ nodes $\boldsymbol{q} = [\boldsymbol{q}_1^\top \cdots \boldsymbol{q}_{12}^\top]^\top \in \mathbb{R}^{36}$, six bars connecting node pairs (i, j) and 24 outer cables; the payload core is a point mass m_c suspended by 12 cables. Bar k of length l_k, mass m_k and radius r_k is a uniform cylinder, which gives the consistent 2×2 block mass matrix (times I_3)

$$\boldsymbol{M}_k = \begin{bmatrix} a_k + m_i & b_k \\ b_k & a_k + m_j \end{bmatrix},\ a_k = \tfrac{m_k}{4} + \tfrac{J_k}{l_k^2},\ b_k = \tfrac{m_k}{4} - \tfrac{J_k}{l_k^2}, \tag{1}$$

with $J_k = \frac{m_k}{12}(3r_k^2 + l_k^2)$ and node masses m_i. Because the bars of a six-bar tensegrity are node-disjoint, $\boldsymbol{M}$ is block diagonal and $\boldsymbol{M}^{-1}$ is available in closed form. The bar-length constraints and their Jacobian are

$$R_k(\boldsymbol{q}) = \|\boldsymbol{b}_k\|^2 - l_k^2 = 0, \quad \boldsymbol{b}_k = \boldsymbol{q}_j - \boldsymbol{q}_i, \quad \boldsymbol{R}_q = \partial \boldsymbol{R}/\partial \boldsymbol{q}. \tag{2}$$

B. Equations of motion

With $\boldsymbol{F}$ collecting gravity, cable, damping, contact and payload forces, the index-1 form of the DAE $\boldsymbol{M}\ddot{\boldsymbol{q}} = \boldsymbol{F} + \boldsymbol{R}_q^\top \boldsymbol{\lambda}$ yields the multipliers in closed form,

$$\boldsymbol{\lambda} = -\big(\boldsymbol{R}_q \boldsymbol{M}^{-1} \boldsymbol{R}_q^\top\big)^{-1} \big(\boldsymbol{\xi}_2 + \boldsymbol{R}_q \boldsymbol{M}^{-1} \boldsymbol{F}\big), \quad \xi_{2,k} = 2\|\dot{\boldsymbol{b}}_k\|^2, \tag{3}$$

and the bar compression $P_k = 2\lambda_k l_k$. The 6×6 system in (3) is solved directly at every step ($\Delta t = 2\,\text{ms}$, semi-implicit Euler).

Algorithm 1 Minimum-norm constraint and energy correction

Require: $\boldsymbol{q}, \dot{\boldsymbol{q}}$ after the step; ledger target $E^\star$

$\delta W \leftarrow 0$
for $it = 1, \ldots, 10$ **do**
  $\boldsymbol{r} \leftarrow [\boldsymbol{R};\ \dot{\boldsymbol{R}};\ E + V_c - (E^\star + \delta W)]$
  if $\|\boldsymbol{r}\|_\infty < \gamma$ **then break**
  end if
  $\boldsymbol{A} \leftarrow \partial \boldsymbol{r}/\partial[\boldsymbol{q}; \dot{\boldsymbol{q}}; \delta W]$
  $\boldsymbol{d} \leftarrow -\boldsymbol{A}^\top (\boldsymbol{A}\boldsymbol{A}^\top)^{-1} \boldsymbol{r}$ ▷ minimum-norm step
  $[\boldsymbol{q}; \dot{\boldsymbol{q}}; \delta W] \leftarrow [\boldsymbol{q}; \dot{\boldsymbol{q}}; \delta W] + \boldsymbol{d}$
end for
book δW in the ledger (category *corrector*)

C. Constraint and energy correction

Integration drift is removed after every step by the minimum-norm corrector of [7] (Algorithm 1). The residual vector stacks the position constraints, the velocity constraints $\dot{R}_k = 2\boldsymbol{b}_k^\top \dot{\boldsymbol{b}}_k$ and, optionally, the energy row $E(\boldsymbol{q}, \dot{\boldsymbol{q}}) - E^\star$, where $E^\star$ is the energy predicted by the work ledger. As in [7] the scalar correction work δW is an additional unknown, so the minimum-norm step distributes the energy correction between the state and an explicitly booked term instead of hiding it. Our extension is the energy function: besides kinetic and gravitational energy it contains the contact potential V_c (below), whose gradient enters the energy row of the Jacobian. The energy row is only enforced while $|E - E^\star| \le \max(0.5\,\text{J}, 0.02|E|)$, so that a gross modelling error is reported by the ledger instead of being hidden by the corrector. Note that with the energy row active the post-correction residual is zero *by construction*; the informative quantity is the pre-correction drift defined below.

D. Passive contact

For a node at penetration $p = \max(0, h(x, y) - z)$ into terrain h, the normal force is the Hunt–Crossley law

$$\boldsymbol{f}_n = k\, p^{3/2}(1 + \alpha \dot{p})\, \hat{\boldsymbol{n}}, \qquad \hat{\boldsymbol{n}} \propto (-h_x, -h_y, 1), \tag{4}$$

whose elastic part is conservative with potential $V_c = \frac{2}{5} k\, p^{5/2}$ and whose damping part is dissipative for every $\dot{p}$. Tangential friction is a regularized Coulomb law $\boldsymbol{f}_t = -\mu \|\boldsymbol{f}_n\|\, \boldsymbol{v}_t / \sqrt{\|\boldsymbol{v}_t\|^2 + \epsilon^2}$, which also only dissipates. No clamps, restitution impulses or velocity projections are used, so the contact cannot inject energy.

E. Work ledger and admissibility certificate

The ledger stores $E = T + V_g + V_c$ and, for every force category c (outer cables, payload cables, contact, drag, disturbance, …), the trapezoidal work $W_c \mathrel{+}= \boldsymbol{F}_c^\top (\dot{\boldsymbol{q}}_k + \dot{\boldsymbol{q}}_{k+1}) \Delta t/2$. Position or velocity projections are booked as the energy change they cause, and parameter changes as the jump of E. The certificate is

$$\varepsilon(t) = E(t) - E_0 - \textstyle\sum_c W_c(t), \tag{5}$$

evaluated after the corrector, and the per-step *drift*

$$\delta_k = \varepsilon_k^- - \varepsilon_{k-1}, \tag{6}$$

where ε_k^- is the residual after step k but before the energy row is applied. δ_k is the energy that step k created or destroyed without a booked cause. For a correct model it is discretization error and must shrink with Δt; an unbooked force or projection would not. We further report the *numerical work* $|W_{\mathrm{num}}|$ (energy changed by constraint projections and by the corrector's δW) and the *aid share*, the fraction of the energy input supplied by non-physical categories (assist torques, virtual forces). We call a run *admissible* if the aid share is zero, $\max_k |\delta_k|$ and $|W_{\mathrm{num}}|$ are small compared with the actuator work, and δ converges with Δt.

IV. Sensitivity-Gradient Tipping

A. Quasi-static sensitivity

Between tips, the rover rests on a support face $\mathcal{F}$ of three nodes that stick to the ground. Augment $\boldsymbol{q}$ with the core position ($n' = 13$ nodes) and linearize the static equilibrium with respect to the cable rest-length offsets $\boldsymbol{u} \in \mathbb{R}^{36}$ ($u > 0$ shortens):

$$\begin{bmatrix} \boldsymbol{K} & \boldsymbol{A}^\top \\ \boldsymbol{A} & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \delta \boldsymbol{q} \\ \delta \boldsymbol{\mu} \end{bmatrix} = \begin{bmatrix} \boldsymbol{B} \\ \boldsymbol{0} \end{bmatrix} \delta \boldsymbol{u}, \quad \boldsymbol{A} = \begin{bmatrix} \boldsymbol{R}_q \\ \boldsymbol{E}_{\mathcal{F}} \end{bmatrix}, \tag{7}$$

where $\boldsymbol{E}_{\mathcal{F}}$ pins the support nodes and $\boldsymbol{B}$ contains $\pm k\,\boldsymbol{e}$ for every taut cable. The tangent stiffness assembles, for a taut cable with axial stiffness k, tension t, length L and direction $\boldsymbol{e}$, the element $k\boldsymbol{e}\boldsymbol{e}^\top + \frac{t}{L}(\boldsymbol{I} - \boldsymbol{e}\boldsymbol{e}^\top)$ and, for a bar under compression P, the *negative* geometric stiffness $-\frac{P}{L}(\boldsymbol{I} - \boldsymbol{e}\boldsymbol{e}^\top)$; axial bar rigidity is already a constraint. A small relative Tikhonov term keeps the KKT matrix regular. One LU factorization of (7) gives all 36 columns of the COM sensitivity $\boldsymbol{S} = \partial \mathbf{c}/\partial \boldsymbol{u} = \frac{1}{m_{\mathrm{tot}}} \sum_i m_i\, \partial \boldsymbol{q}_i / \partial \boldsymbol{u}$.

B. Edge selection and command

For each edge e of $\mathcal{F}$ with outward horizontal normal $\boldsymbol{n}_e$ and midpoint $\boldsymbol{p}_e$, the tipping margin is $m_e = -(\boldsymbol{c} - \boldsymbol{p}_e)^\top \boldsymbol{n}_e$ and the gradient of the COM shift across the edge is $\boldsymbol{g}_e = \boldsymbol{S}_{xy}^\top \boldsymbol{n}_e$. Under box limits $u_k \in [\underline{u}_k, \bar{u}_k]$ the achievable shift is $s_e = \sum_k \max(g_{e,k}\bar{u}_k,\, g_{e,k}\underline{u}_k)$. Among edges whose normal is aligned with the travel direction ($\boldsymbol{n}_e^\top \boldsymbol{d} \geq 0.25$), SGT locks for the whole cycle the edge with the largest reachable margin $s_e - m_e$ and commands

$$u_k = \bar{u}_k\, \mathrm{clip}\big(g_{e,k}/\|\boldsymbol{g}_e\|_\infty,\, -\rho,\, 1\big), \tag{8}$$

separately normalized for shell and payload cables; ρ limits cable lengthening. A four-state machine (Settle→Shift→Tip→Release) re-plans at 20 Hz until the support face changes, releases all cables and waits for the shell to settle (with a timeout) before the next cycle. SGT uses only cable rest lengths; the simulator runs with all assists disabled.

Design rationale. A support-face primitive contracts a fixed cable set; most of its effect is absorbed by elastic deformation that does not move the COM over an edge. The sensitivity of the *constrained* system instead tells which cables move the COM in the required direction, and the compressed bars soften the structure precisely in the bending directions that tipping needs. The ablations in Section VI test which of these ingredients matter.

V. Experimental Protocol

Platform. Shell radius 0.5 m (bar length 0.851 m), 12 nodes of 0.2 kg, six bars of 0.3 kg, payload 1.6 kg (5.8 kg in total); cable stiffness 1200 N/m (payload cables 1600 N/m). Every cable-only method uses the same actuator box: stroke 0.15 m (30 % of the 0.49 m rest length) for shell and payload cables and 0.20 m/s spool speed. Physics runs at 500 Hz, controllers at 20 Hz. Unless stated otherwise all runs use the DAE core with Algorithm 1, passive contact and zero aids, on terrain Level 1 for 20 s.

Methods. SGT (proposed). *Support-face heuristic*: the fixed contraction primitive around the current face. *LQR, iLQR, QP-MPC*: the controllers of the original code base, which solve a two-state reduced-order rolling model (speed along the goal, height deviation) for a normalized effort $u \in [-1, 1]$ that scales the same support-face primitive—Riccati backward pass, nonlinear iLQR with line search, and box-constrained projected-gradient MPC, respectively. *Legacy+assist*†: the original penalty+SHAKE engine with roll couple, traction, grip and corridor aids; reported as a reference only, because it is not admissible.

Scenarios. Lateral impulse $\Delta v = 0.5$ m/s at $t = 6$ s; step push 10 N for 2 s; Gaussian, Laplace and outlier (2 %, 10×) noise on the measured state; cable stiffness −20 %, friction −40 %, payload mass +30 %; actuation delay 100 ms.

Metrics. Mean forward speed $\bar{v}$; the cross-track error $e(t) = x_c(t) - x_c(0)$ against the straight reference heading, reported as RMS and maximum $|e|$; success ($\Delta y \geq 1$ m in 20 s); recovery time after a disturbance; control effort $\int \boldsymbol{u}^\top \boldsymbol{u}\, dt$; actuator work and cost of transport $\mathrm{CoT} = W_{\mathrm{act}}/(mg\,\Delta y)$; controller wall time (mean and worst); $\max|\varepsilon|$, $\max_k |\delta_k|$, $\sum_k |\delta_k|$, $|W_{\mathrm{num}}|$, maximal bar-length error and aid share. Statistics are mean ± std over seeds 1–5 (nominal) or 1–3 (other groups). A seed selects the procedural terrain outside a flat 1.5 m launch pad and the sensor-noise stream; methods that never leave the pad therefore produce identical runs (zero spread).

VI. Results

A. Physical consistency

Table II and Fig. 1 compare the cores under the same SGT command sequence. Three observations stand out. *(i) Rigidity.* The DAE core keeps the bar lengths at 2.3×10^{-11} m; the penalty+SHAKE core reaches 1.3×10^{-7} m and, with the interactive aids, 1.9×10^{-4} m. *(ii) Hidden energy exchange.* To keep the bars rigid, the penalty+SHAKE core removes 6.8×10^{3} J by position projection while its cables do a comparable amount of work, i.e. the cables mostly pump energy into the penalty bars that the projection then discards. The DAE core changes only 7.3×10^{-3} J numerically. With the original locomotion aids, 50 % of the energy input comes from non-physical forces, which explains its high speed (0.522 m/s) in Table I. *(iii) Drift.* Before correction, both cores show a per-step energy mismatch that is first-order discretization error: for seed 1,

TABLE I

NOMINAL CABLE-ONLY LOCOMOTION ON LEVEL 1 (20 S, 5 SEEDS, MEAN ± STD). ALL METHODS USE THE LAGRANGIAN DAE CORE, PASSIVE CONTACT AND IDENTICAL ACTUATOR LIMITS.

Method	$\bar{v}$ (m/s)	RMSE (m)	max $\lvert e\rvert$ (m)	$\int u^\top u\,dt$	CoT	t_{solve} (ms)	Succ. (%)	Aid (%)
SGT (proposed)	$\mathbf{0.091 \pm 0.026}$	0.253 ± 0.029	0.473 ± 0.000	$\mathbf{0.334 \pm 0.170}$	$\mathbf{11.4 \pm 6.6}$	0.68 ± 0.10	**100**	0
Heuristic	0.004	0.083	0.160	2.059	142.4	$\mathbf{0.47 \pm 0.25}$	0	0
LQR	0.002	0.129	0.163	2.529	–	0.65 ± 0.06	0	0
iLQR	0.005	0.081	0.168	1.757 ± 0.000	111.8	0.90 ± 0.41	0	0
QP-MPC	0.004	0.084	0.175	1.859 ± 0.000	132.0	0.82 ± 0.05	0	0
Legacy+assist†	0.522 ± 0.023	0.006 ± 0.002	0.017 ± 0.006	0.636 ± 0.060	1.1 ± 0.1	0.15 ± 0.02	100	50

†Legacy engine with external roll couple, traction, grip and corridor aids; not physically admissible and excluded from the bold "best" marks. Success: $\Delta y \geq 1$ m in 20 s. Aid: share of the energy input supplied by non-physical aids. Cross-track errors are not marked because they are only comparable between methods that travel a similar distance.

TABLE II

PHYSICAL CONSISTENCY OF THE SIMULATOR UNDER THE SGT GAIT (LEVEL 1, 20 S). $\varepsilon = E - E_0 - \sum W$ IS THE ENERGY NO BOOKED WORK EXPLAINS *after* CORRECTION (ENFORCED BY THE ENERGY ROW); δ_k IS THE SAME QUANTITY FOR STEP k *before* CORRECTION, I.E. THE PER-STEP BOUND ON ANY UNBOOKED ENERGY SOURCE; $|W_{\mathrm{num}}|$ IS THE ENERGY CHANGED BY CONSTRAINT PROJECTIONS AND BY THE CORRECTOR (BOOKED, BUT NUMERICAL). SGT ON LEVEL 1, 20 S; †THE ORIGINAL ENGINE WITH ITS LOCOMOTION AIDS AND SUPPORT-FACE GAIT, FOR REFERENCE.

Method	max $\lvert\varepsilon\rvert$ (J)	max $\lvert\delta_k\rvert$ (J)	$\sum \lvert\delta_k\rvert$ (J)	bar err. (m)	$\lvert W_{\mathrm{num}}\rvert$ (J)	$\bar{v}$ (m/s)	Aid (%)
DAE + Alg. 1, energy row	1.0×10^{-9}	7.6×10^{-2}	1.7×10^{1}	2.3×10^{-11}	7.3×10^{-3}	0.091 ± 0.026	0
DAE + Alg. 1, constr. only	1.5×10^{1}	7.1×10^{0}	3.3×10^{1}	5.9×10^{-11}	3.9×10^{1}	0.104 ± 0.029	0
Penalty + SHAKE (legacy)	1.1×10^{1}	1.1×10^{-3}	1.1×10^{1}	1.3×10^{-7}	6.8×10^{3}	0.026	0
Legacy + assists†	1.1×10^{1}	1.2×10^{-3}	1.1×10^{1}	1.9×10^{-4}	1.5×10^{4}	0.522 ± 0.023	50

TABLE III

ROBUSTNESS: DISTURBANCES, SENSOR NOISE, PLANT MISMATCH AND ACTUATOR DELAY (LEVEL 1, 20 S, 3 SEEDS).

Scenario	SGT $\bar{v}$ (m/s)	Heur. $\bar{v}$ (m/s)	SGT RMSE (m)	t_{rec} (s)	SGT succ. (%)
Nominal	0.080 ± 0.029	0.004	0.258 ± 0.040	–	100
Lateral impulse 0.5 m/s	0.094 ± 0.017	0.004	0.266 ± 0.011	0.09	100
Step push 10 N, 2 s	0.085 ± 0.006	-0.001	0.308 ± 0.035	2.00	100
Gaussian sensor noise	0.065 ± 0.016	0.001 ± 0.002	0.219 ± 0.030	–	67
Laplace sensor noise	0.026 ± 0.016	-0.001 ± 0.004	0.158 ± 0.036	–	0
Sensor outliers (2 %, 10x)	0.034 ± 0.023	0.002 ± 0.002	0.167 ± 0.052	–	33
Cable stiffness -20 %	0.057 ± 0.002	0.002	0.226 ± 0.004	–	100
Friction -40 %	0.104 ± 0.024	0.002	0.224 ± 0.051	–	100
Payload mass +30 %	0.083 ± 0.016	0.002	0.249 ± 0.022	–	100
Actuator delay 100 ms	0.050 ± 0.003	0.004	0.192 ± 0.002	–	67

TABLE IV

ABLATION OF THE SGT GAIT AND OF THE PHYSICS CORE (LEVEL 1, 20 S, 3 SEEDS).

Variant	$\bar{v}$ (m/s)	tips	RMSE (m)	CoT	Succ. (%)
SGT (full)	0.080 ± 0.029	8.7 ± 3.5	0.258 ± 0.040	13.4 ± 8.4	**100**
w/o payload actuation	0.076 ± 0.005	8.7 ± 0.6	0.242 ± 0.006	7.7 ± 0.0	**100**
w/o feasibility edge choice	0.102 ± 0.034	10.3 ± 0.6	0.260 ± 0.041	8.9 ± 3.8	**100**
w/o bar geometric stiffness	0.056 ± 0.010	5.7 ± 1.2	0.194 ± 0.001	23.5 ± 4.3	67
sparse: 8 cables	0.085 ± 0.002	9.3 ± 0.6	0.286 ± 0.025	$\mathbf{5.6 \pm 0.6}$	**100**
DAE w/o energy row	$\mathbf{0.104 \pm 0.029}$	10.3 ± 2.1	0.242 ± 0.026	9.4 ± 5.1	**100**
legacy SHAKE core	0.026	**14.0**	**0.043**	49.1	0

TABLE V

TERRAIN GENERALIZATION (3 SEEDS PER CELL).

Terrain / method	$\bar{v}$ (m/s)	RMSE (m)	Succ. (%)	max $\lvert\varepsilon\rvert$ (J)
Level 1: SGT	0.080 ± 0.029	0.258 ± 0.040	100	1.0×10^{-9}
Level 1: Heuristic	0.004	0.083	0	9.9×10^{-10}
Level 4: SGT	0.074 ± 0.019	0.275 ± 0.044	100	1.0×10^{-9}
Level 4: Heuristic	0.004	0.083	0	9.9×10^{-10}
Level 7: SGT	0.087 ± 0.020	0.229 ± 0.008	100	1.5×10^{-2}
Level 7: Heuristic	0.004	0.083	0	9.9×10^{-10}

halving Δt from 2 to 1 ms reduces the accumulated $\sum_k |\delta_k|$ from 15.4 to 7.6 J for the DAE core and from 11.0 to 5.5 J for penalty+SHAKE, and the largest single-step mismatch of the DAE core (at the initial landing) from 7.6×10^{-2} to 1.9×10^{-2} J. The energy row removes this drift every step, so ε stays at the solver tolerance (1.0×10^{-9} J). Without the energy row the residual accumulates to 1.5×10^{1} J. We stress that the energy row corrects an integration error of known order; it does not by itself prove that the model is complete—that is the role of δ_k, which would expose an unbooked force as a non-convergent drift. On rugged Level 7 the corrector occasionally reaches its 10-iteration cap during contact transitions, leaving a transient residual of up to 4×10^{-2} J (0.2 % of E) that is removed in the following steps (Table V).

B. Nominal locomotion

With identical actuator limits and all aids disabled, SGT advances 1.81 m in 20 s (0.091 ± 0.026 m/s, 100 % success; Table I, Fig. 2), about 25 times the speed of the support-face heuristic. The LQR, iLQR and QP-MPC schedulers optimize a scalar effort over the same support-face primitive and therefore inherit its limitation: none of them leaves the launch pad (at most 0.10 m), and their cost of transport is an order of magnitude higher. This is a statement about the primitive, which cannot shift the COM of the 5.8 kg system over an edge within a 0.15 m stroke, rather than about the optimizers. SGT plans in 0.68 ms on average (worst 17.4 ms) in single-threaded JavaScript, well inside its 50 ms control period. Its main weakness is heading: the greedy per-cycle edge choice produces a zig-zag track with a cross-track RMSE of 0.253 m (Fig. 3).

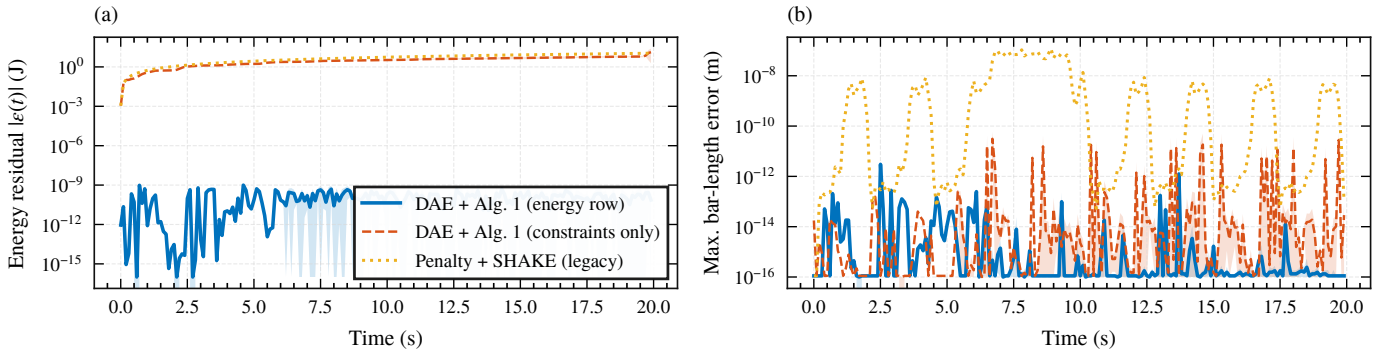


Fig. 1. Physical consistency under the SGT gait (Level 1, mean $\pm 1\sigma$ shaded). (a) Ledger residual $|\varepsilon(t)|$ after correction (held at the solver tolerance by the energy row; it accumulates without it). (b) Maximal bar-length error. Solid: DAE with energy row; dashed: DAE with constraint rows only; dotted: legacy penalty+SHAKE core.

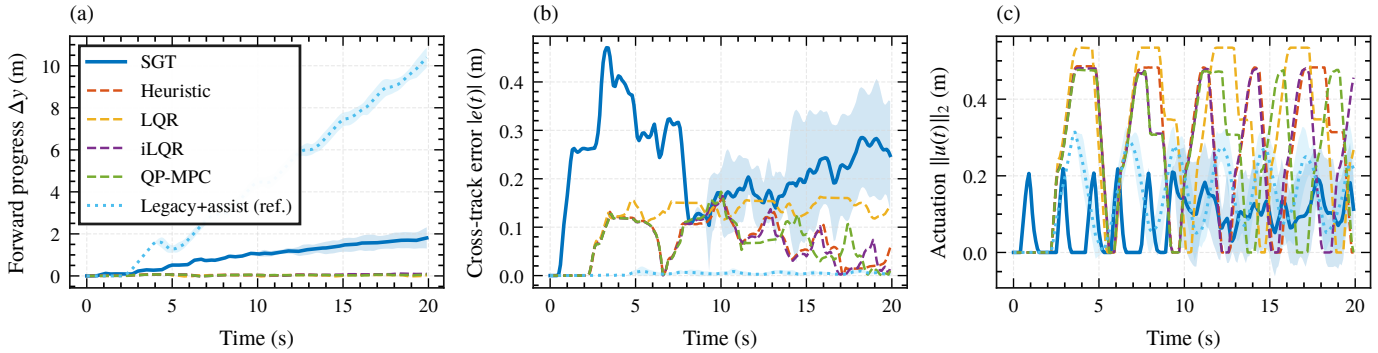


Fig. 2. Nominal cable-only locomotion (5 seeds, mean $\pm 1\sigma$). (a) Forward progress. (b) Cross-track error. (c) Actuation norm. Solid: proposed; dashed: baselines; dotted: inadmissible reference.

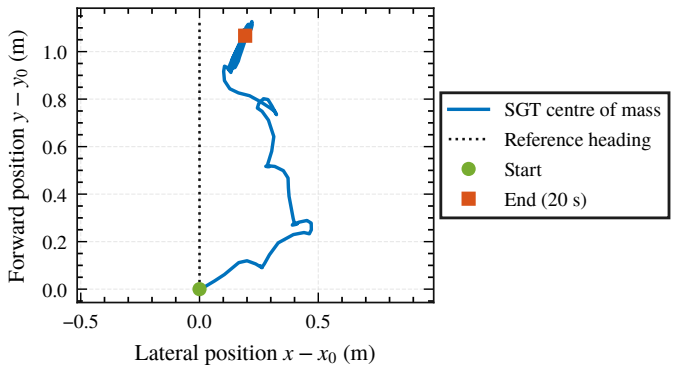


Fig. 3. Ground track of the COM under SGT (seed 1, 20 s).

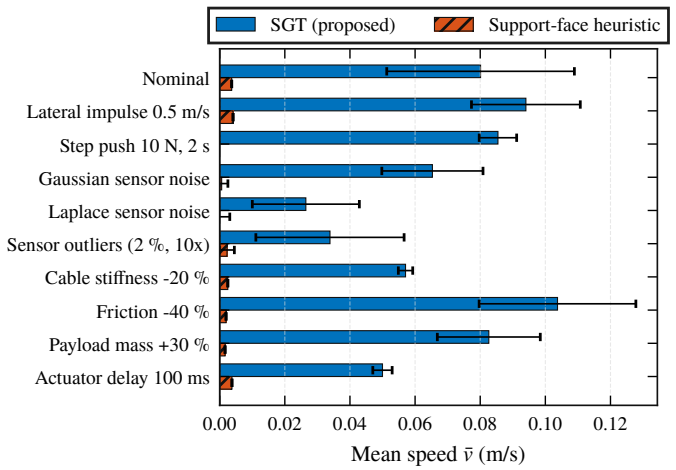


Fig. 4. Mean speed under disturbances, sensor noise, plant mismatch and delay (3 seeds, error bars $\pm 1\sigma$).

C. Robustness

Table III and Fig. 4 summarize the stress tests (3 seeds each). SGT keeps rolling after a 0.5 m/s lateral impulse (0.094 m/s) and under a 10 N, 2 s push (0.085 m/s), and is insensitive to $-40\,\%$ friction (0.104 m/s) and $+30\,\%$ payload mass (0.083 m/s). Softer cables ($-20\,\%$) slow it to 0.057 m/s because the same stroke produces smaller COM shifts, and a 100 ms actuation delay to 0.050 m/s (67 % success). Measurement noise is the main weakness: Gaussian noise reduces the speed to 0.065 m/s, while heavy-tailed noise is damaging—Laplace noise to 0.026 m/s (0 % success) and 2 % outliers to 0.034 m/s. SGT uses the measured node positions both to detect the support face and to assemble $\boldsymbol{K}$, so a single corrupted node can trigger a wrong face or edge; a robust state estimator in front of the planner is required before hardware use.

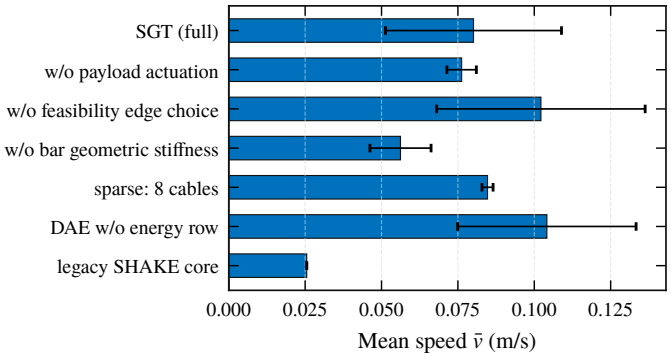


Fig. 5. Ablation of SGT components and of the physics core (3 seeds).

D. Ablation and terrain

Table IV isolates the ingredients of SGT (3 seeds). Removing the bars' geometric stiffness is the most harmful change: speed drops from 0.080 to 0.056 m/s, success to 67 % and the cost of transport nearly doubles (23.5 vs. 13.4), confirming that the softening of the compressed bars is what the sensitivity has to capture. The other design choices did not help in our tests: without payload actuation the speed is essentially unchanged (0.076 m/s) at a lower cost of transport (7.7); restricting the command to the 8 most sensitive cables keeps the speed (0.085 m/s) and gives the lowest cost of transport (5.6); and choosing the edge by alignment with the travel direction instead of by reachable margin was not worse (0.102 ± 0.034 m/s). With three seeds these differences are within one standard deviation, so we report the margin-based rule as specified before the experiments and leave the sparse, payload-free variant as the more efficient configuration. Run on the penalty+SHAKE core without aids, the same controller reaches only 0.026 m/s: the projection that absorbs the cable work also absorbs the deformation that SGT relies on. Across terrains (Table V) SGT keeps 0.074 m/s on Level 4 and 0.087 m/s on Level 7 with 100 % success.

VII. Limitations

All results are in simulation. The contact parameters (k, α, μ) and cable stiffnesses were not identified on hardware, so absolute speeds will not transfer directly; what the ledger provides is a bound on the energy contributed by the integrator, projections and aids, not a validation of the model against reality. The time-step study uses a single seed. SGT is sensitive to heavy-tailed measurement noise (Section VI) and needs a robust estimator before hardware use. SGT is quasi-static: it ignores inertia during the tip and re-plans only at 20 Hz, which limits speed to the order of $0.1\,\mathrm{m/s}$. Edge selection is greedy per cycle, which causes the zig-zag ground track and cross-track error visible in Fig. 3; a heading-aware selection over several faces is a natural extension. The baselines inherit a reduced-order model and the support-face primitive from the original code base; stronger whole-body optimal controllers may close part of the gap, and the released protocol is intended to make such comparisons easy.

VIII. Conclusion

We showed how to make tensegrity-rover simulation *auditable*: a non-minimal Lagrangian DAE with exact multipliers, a minimum-norm constraint-and-energy corrector, passive contact with a potential and a work ledger that measures the unexplained energy of every step before it is corrected. The ledger shows that this drift is first-order discretization error in the proposed core, and it exposes the kilojoule-scale projection exchange and the non-physical aids of a conventional penalty+projection engine. On this basis, the sensitivity-gradient tipping gait achieves cable-only rolling of a six-bar rover with a heavy payload where reduced-order effort schedulers over a fixed primitive do not; the bars' geometric stiffness is the ingredient of the sensitivity that matters most. Code, run files and the scripts that generate every table and figure are released with the paper.

Reproducibility

`npm run experiments` executes the full protocol (114 runs, resumable), `npm run paper:data` regenerates `tables.tex`, `numbers.tex` and all figures, and `node tools/drift_check.mjs` repeats the time-step study; the same protocol runs in the browser (*Comparative benchmark* panel).

References

[1] A. P. Sabelhaus, J. Bruce, K. Caluwaerts, P. Manovi, R. F. Firoozi, S. Dobi, A. M. Agogino, and V. SunSpiral, "System design and locomotion of SUPERball, an untethered tensegrity robot," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2015, pp. 2867–2873.

[2] Z. Littlefield, D. Surovik, M. Vespignani, J. Bruce, W. Wang, and K. E. Bekris, "Kinodynamic planning for spherical tensegrity locomotion with effective gait primitives," *The International Journal of Robotics Research*, 2019.

[3] B. T. Mirletz, I.-W. Park, R. D. Quinn, and V. SunSpiral, "Towards bridging the reality gap between tensegrity simulation and robotic hardware," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2015, pp. 5357–5363.

[4] M. Zhang, X. Geng, J. Bruce, K. Caluwaerts, M. Vespignani, V. SunSpiral, P. Abbeel, and S. Levine, "Deep reinforcement learning for tensegrity robot locomotion," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2017.

[5] J.-P. Ryckaert, G. Ciccotti, and H. J. C. Berendsen, "Numerical integration of the Cartesian equations of motion of a system with constraints: molecular dynamics of n-alkanes," *Journal of Computational Physics*, vol. 23, no. 3, pp. 327–341, 1977.

[6] H. C. Andersen, "Rattle: A "velocity" version of the Shake algorithm for molecular dynamics calculations," *Journal of Computational Physics*, vol. 52, no. 1, pp. 24–34, 1983.

[7] TODO: authors of the repository preprint, "TODO: exact title – lagrangian tensegrity dynamics in non-minimal cartesian coordinates with direct constraint and energy correction," TODO, preprint (repository file tensegrityDynamicspreprint.pdf); the scanned copy lacks its title page – fill in before submission.

[8] K. H. Hunt and F. R. E. Crossley, "Coefficient of restitution interpreted as damping in vibroimpact," *Journal of Applied Mechanics*, vol. 42, no. 2, pp. 440–445, 1975.

[9] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *Proc. IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2012, pp. 5026–5033.

[10] E. Coumans, "Bullet physics simulation," ACM SIGGRAPH 2015 Courses, 2015.

[11] J. Baumgarte, "Stabilization of constraints and integrals of motion in dynamical systems," *Computer Methods in Applied Mechanics and Engineering*, vol. 1, no. 1, pp. 1–16, 1972.

[12] E. Hairer, C. Lubich, and G. Wanner, *Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations*, 2nd ed. Springer, 2006.
[13] R. E. Skelton and M. C. de Oliveira, *Tensegrity Systems*. Springer, 2009.